DigiPen
INSTITUTE OF TECHNOLOGY

CS 372 Machine Learning I
Project Presentation Guidelines

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Outlines

- Project Presentation & logistics.

Deliverables

- Project Proposal Submission 10%
  - ✓Reflection Video on proposal presentation 5%
  - ✓Proposal presentation 5%

- Project Final Submission 90%
  - A successful project implementation + Demo (50%)
  - A project presentation (20%)
  - Reflection Video on final presentation (20 %)

Proposal Presentation

When:

- Friday March 28th, 2025, during class.

Where:

- W-204.

Who:

- Every student.

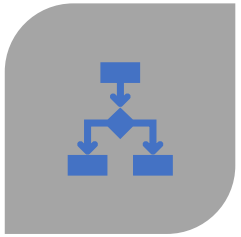
Duration:

- Project Proposal ~5 minutes.
- Final Presentation ~ 20 minutes.

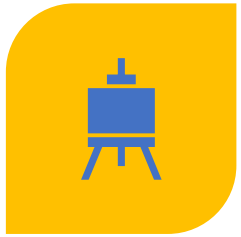
Presentation Preparation



APPROPRIATE THEME.



CONTENT



FONT AND FORMAT CONSISTENCY.



GRAMMAR AND SPELLING.



TIME OF THE PRESENTATION.



PRACTICE

Presentation Main Topics

- Title Slide.
- Outline.
- Background.
- Motivation & problem statement.
- Project Overview.
- References.
- Demo.

Presentation

Title Slide (first slide), includes:

- Project Title.
- Student name.
- Course name.
- Introduce yourself.

Good Project title

- general idea of the project.
  - Example: Spam filtering using Machine Learning is **too general**.
- simple and straightforward.
- memorable.
- not too long.

Presentation Main Topics

- Title Slide.
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Presentation Outline

- Clearly state the outline of your presentation in the second slide.

Presentation Main Topics

- Title Slide.
- Outline.
- Background.
- Motivation & problem statement.
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Background and topic Overview

- High level introduction.
- Big picture of the problem: what, why, how.
- Graphics and visualization.

Presentation Main Topics

- Title Slide.
- Outline.
- Background.
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Motivation and Problem Statement

- What are the problems that you are trying to address?

- Why would machine learning help solving this problem?

Presentation Main Topics

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Project Overview

- Objectives & goals
- Data Overview.
- Machine Learning Algorithms.
- Experiments & Discussion.
- Challenges.
- Future work.

Objectives & Goals

- Clearly state the objectives of your project.

Data Overview

- Dataset Collection.

- Exploratory Data Analysis.
  - Dataset statistics and examples of the dataset used.
  - Visualization.

- You can find datasets at
    - Google Dataset Search
    - Kaggle
    - UCI Machine Learning Repository
    - CMU Libraries
    - PMC Text Mining Collections
    - ImageNet

Machine Learning Algorithms

- Machine Learning Algorithms you use in the project.

Challenges

- Explain why this is a challenging Problem.
- Discuss challenges you have encountered so far. How you handled them and why.
- Challenges you think you will face.

Experiments & Discussion

- Explain the experiments that were carried out.
- Explain the result of your experiments.
  - Discuss what worked and what not.
- Discuss some future work.

Future timeline (Project Proposal Slide)

- Progress you have made so far.

- Plan your future objectives.

Presentation Main Topics

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References

- Include research papers you have read so far.
- Be consistent.
- There various citations style:
  - IEEE (Institute of Electrical and Electronics Engineers) is most common.
    - https://ieeeauthorcenter.ieee.org/wp-content/uploads/IEEE-Reference-Guide.pdf

Presentation Main Topics

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Demo (Final Presentation)

- Show us a Demo of your project ☺

Questions!